4.5.6 Representing images, sound and other data

4.5.6.1 Bit patterns, images, sound and other data

Content	Additional information
Describe how bit patterns may represent other forms of data, including graphics and sound.	

4.5.6.2 Analogue and digital

Content	Additional information
Understand the difference between analogue and digital: • data • signals.	

4.5.6.3 Analogue/digital conversion

Content	Additional information
Describe the principles of operation of: • an analogue to digital converter (ADC) • a digital to analogue converter (DAC).	
Know that ADCs are used with analogue sensors.	

Content	Additional information
Know that the most common use for a DAC is to convert a digital audio signal to an analogue signal.	

4.5.6.4 Bitmapped graphics

Content	Additional information
Explain how bitmaps are represented.	
Explain the following for bitmaps: • resolution • colour depth • size in pixels.	The size of an image is also alternatively sometimes described as the resolution of an image. Size of an image in pixels is width of image in pixels x height of image in pixels. Resolution is expressed as number of dots per inch where a dot is a pixel. Colour depth = number of bits stored for each pixel.
Calculate storage requirements for bitmapped images and be aware that bitmap image files may also contain metadata.	Ignoring metadata, *storage requirements* = *size in pixels* x *colour depth* where size in pixels is *width in pixels* x *height in pixels*.
Be familiar with typical metadata.	eg width, height, colour depth.

4.5.6.5 Vector graphics

Content	Additional information
Explain how vector graphics represents images using lists of objects.	The properties of each geometric object/shape in the vector graphic image are stored as a list.
Give examples of typical properties of objects.	
Use vector graphic primitives to create a simple vector graphic.	

4.5.6.6 Vector graphics versus bitmapped graphics

Content	Additional information
Compare the vector graphics approach with the bitmapped graphics approach and understand the advantages and disadvantages of each.	
Be aware of appropriate uses of each approach.	

4.5.6.7 Digital representation of sound

Content	Additional information
Describe the digital representation of sound in terms of: • sample resolution • sampling rate and the Nyquist theorem.	
Calculate sound sample sizes in bytes.	

4.5.6.8 Musical Instrument Digital Interface (MIDI)

Content	Additional information
Describe the purpose of MIDI and the use of event messages in MIDI.	
Describe the advantages of using MIDI files for representing music.	

4.5.6.9 Data compression

Content	Additional information
Know why images and sound files are often compressed and that other files, such as text files, can also be compressed.	
Understand the difference between lossless and lossy compression and explain the advantages and disadvantages of each.	
Explain the principles behind the following techniques for lossless compression: • run length encoding (RLE) • dictionary-based methods.	

4.5.6.10 Encryption

Content	Additional information
Understand what is meant by encryption and be able to define it.	Students should be familiar with the terms cipher, plaintext and ciphertext. Caesar and Vernam ciphers are at opposite extremes. One offers perfect security, the other doesn't. Between these two types are ciphers that are computationally secure – see below. Students will be assessed on the two types. Ciphers other than Caesar may be used to assess students' understanding of the principles involved. These will be explained and be similar in terms of computational complexity.
Be familiar with Caesar cipher and be able to apply it to encrypt a plaintext message and decrypt a ciphertext. Be able to explain why it is easily cracked.	
Be familiar with Vernam cipher or one-time pad and be able to apply it to encrypt a plaintext message and decrypt a ciphertext. Explain why Vernam cipher is considered as a cypher with perfect security.	Since the key k is chosen uniformly at random, the ciphertext c is also distributed uniformly. The key k must be used once only. The key k is known as a one-time pad.
Compare Vernam cipher with ciphers that depend on computational security.	Vernam cipher is the only one to have been mathematically proved to be completely secure. The worth of all other ciphers ever devised is based on computational security. In theory, every cryptographic algorithm except for Vernam cipher can be broken, given enough ciphertext and time.